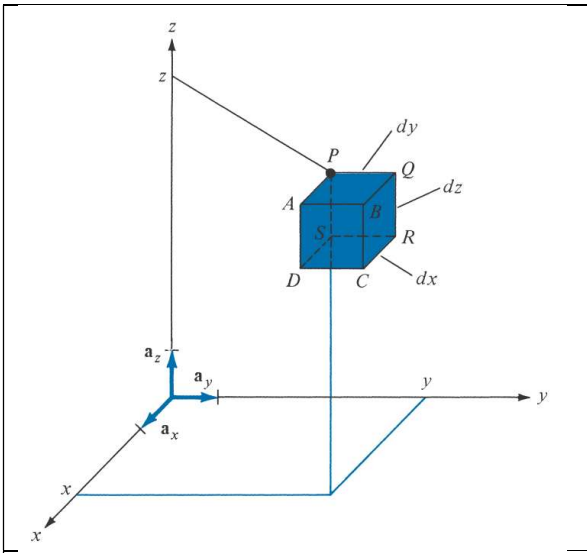
1. POINT AND VECTOR TRANSFORMATION

Point Transformation $\boldsymbol{P(x,y,z) = P(\rho,\phi,z) = P(r,\theta,\phi)}$	
Cartesian to **cylindrical**	$P\begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow P\begin{pmatrix} \rho = \sqrt{x^2+y^2} \\ \phi = tan^{-1}\left(\frac{y}{x}\right) \\ z = z \end{pmatrix}$
Cylindrical to **cartesian**	$P\begin{pmatrix} \rho \\ \phi \\ z \end{pmatrix} \rightarrow P\begin{pmatrix} x = \rho\cos\phi \\ y = \rho\sin\phi \\ z = z \end{pmatrix}$
Cartesian to **spherical**	$P\begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow P\begin{pmatrix} r = \sqrt{x^2+y^2+z^2} \\ \theta = tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right) \\ \phi = tan^{-1}\left(\frac{y}{x}\right) \end{pmatrix}$
Spherical to **cartesian**	$P\begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \rightarrow P\begin{pmatrix} x = rsin\theta\cos\phi \\ y = rsin\theta\sin\phi \\ z = rcos\theta \end{pmatrix}$
Cylindrical to **spherical**	$P\begin{pmatrix} \rho \\ \phi \\ z \end{pmatrix} \rightarrow P\begin{pmatrix} r = \sqrt{\rho^2+z^2} \\ \theta = tan^{-1}\left(\frac{\rho}{z}\right) \\ \phi = \phi \end{pmatrix}$
Spherical to **cylindrical**	$P\begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \rightarrow P\begin{pmatrix} \rho = rsin\theta \\ \phi = \phi \\ z = rcos\theta \end{pmatrix}$

Vector Transformation $\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z} = A_\rho\hat{\rho} + A_\phi\hat{\phi} + A_z\hat{z} = A_r\hat{r} + A_\theta\hat{\theta} + A_\phi\hat{\phi}$	
Cartesian to **cylindrical**	$\vec{A} = \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} = \begin{pmatrix} cos\phi & sin\phi & 0 \\ -sin\phi & cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$
Cylindrical to **cartesian**	$\vec{A} = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} cos\phi & -sin\phi & 0 \\ sin\phi & cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix}$
Cartesian to **spherical**	$\vec{A} = \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} = \begin{pmatrix} sin\theta cos\phi & sin\theta sin\phi & cos\theta \\ cos\theta cos\phi & cos\theta sin\phi & -sin\theta \\ -sin\phi & cos\phi & 0 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$
Spherical to **cartesian**	$\vec{A} = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} sin\theta cos\phi & cos\theta cos\phi & -sin\phi \\ sin\theta sin\phi & cos\theta sin\phi & cos\phi \\ cos\theta & -sin\theta & 0 \end{pmatrix} \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix}$
Cylindrical to **spherical**	$\vec{A} = \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} = \begin{pmatrix} sin\theta & 0 & cos\theta \\ cos\theta & 0 & -sin\theta \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix}$
Spherical to **cylindrical**	$\vec{A} = \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} = \begin{pmatrix} sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \\ cos\theta & -sin\theta & 0 \end{pmatrix} \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix}$

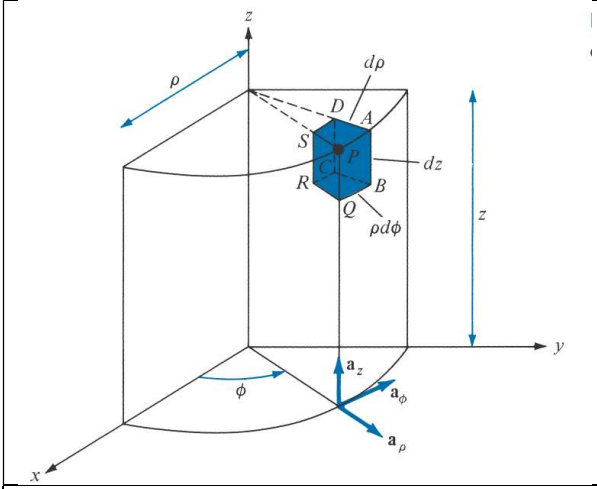
2. DIFFERENTIAL LENGTH, SURFACE AND VOLUME



$$\overrightarrow{dl} = dx\hat{x} + dy\hat{y} + dz\hat{z}$$

$$\overrightarrow{dS} = \begin{cases} \pm dydz\hat{x} \\ \pm dxdz\hat{y} \\ \pm dxdy\hat{z} \end{cases}$$

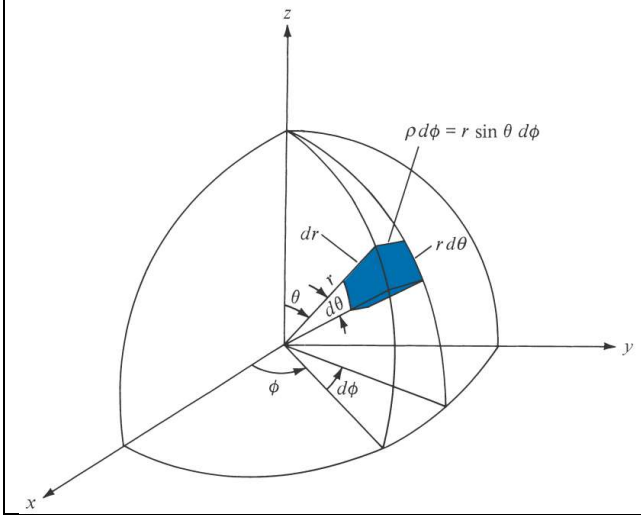
$$dv = dxdydz$$



$$\overrightarrow{dl} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$$

$$\overrightarrow{dS} = \begin{cases} \pm \rho d\phi dz\hat{\rho} \\ \pm d\rho dz\hat{\phi} \\ \pm \rho d\rho d\phi\hat{z} \end{cases}$$

$$dv = \rho d\rho d\phi dz$$



$$\overrightarrow{dl} = dr\hat{r} + rd\theta\hat{\theta} + rsin\theta d\phi\hat{\phi}$$

$$\overrightarrow{dS} = \begin{cases} \pm r^2 \sin\theta\, d\theta d\phi\hat{r} \\ \pm r \sin\theta\, dr d\phi\hat{\theta} \\ \pm r dr d\theta\hat{\phi} \end{cases}$$

$$dv = r^2 \sin\theta\, dr d\theta d\phi$$